

Scaling for Success?

Evaluating the Impact of Canada’s Innovation Policies

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Abstract

This paper evaluates whether Canada’s innovation policies effectively support scale-ups. These are the small group of firms that achieve rapid scale-up growth and drive a disproportionate share of national employment and output. Despite extensive policy attention, there has been little empirical evidence on program impact. We address this gap by linking Statistics Canada’s administrative firm data with records from the Business Innovation and Growth Support database and the Scientific Research and Experimental Development (SR&ED) tax credit program. This is the first study in Canada to use linked administrative data to track scale-ups and federal support at the firm level. We find that scale-ups represent only a small share of firms yet account for large shares of employment and revenue. Most never receive federal support. SR&ED typically occurs before or during scaling, while BIGS funding arrives afterward. Econometric analysis shows that BIGS recipients experience higher growth in employment, revenue, R&D, wages, and exports. The study clarifies how innovation policy interacts with firm growth and where alignment can improve.

Keywords: scale-ups, high-growth firms, innovation policy, administrative data, Canada

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1 Introduction

Over the past two decades, the disruptive effects of new technologies, the climate crisis, and shifting geopolitical relations have underscored the necessity for small open economies to adopt more strategic approaches to supporting high-growth firms. Most often, this takes the form of new industrial policies (Aiginger and Rodrik, 2020). A critical role for the public sector in implementing such policy involves gathering information, targeting high-growth firms across the economy, and effectively allocating public resources to activities that enhance these firms’ ability to expand into global markets. Over the past five decades, Canadian governments have pursued a broad range of innovation policies with little discernible impact on overall innovation performance (Council of Canadian Academies, 2018, 2025; Stewart, 2025). What has been notably lacking to date are detailed empirical assessments of the impact of core policies on the economic performance of the firms they target.

This article addresses that gap by focusing on the small subset of high-performing firms with the potential to drive a disproportionate share of economic growth. In Canada, as in many advanced economies, most firms grow slowly, if at all, while a tiny fraction achieve rapid growth in jobs or revenue (Dixon and Rollin, 2014). These exceptional firms – often termed “scale-ups” or high-growth firms – have drawn increasing attention for their outsized contributions to employment, innovation, and productivity (Coad et al., 2014; Haltiwanger, Jarmin, and Miranda, 2013). For example, in the United Kingdom, just 6 percent of businesses generated about half of the new jobs in the 2000s. Similar patterns hold in Canada: firm growth-rate distributions are highly skewed, with most businesses clustered around zero growth, but a long tail of “gazelles” far outperforming the norm (Nesta, 2013). This heavy-tailed distribution means that scale-ups – though rare – play a critical role in driving net employment creation and economic dynamism (Birch, 1979; Henrekson and Johansson, 2010). Despite their importance, Canadian scale-ups have traditionally received less policy attention than other drivers of economic growth. Historically, Canada’s growth strategy focused heavily on large, often foreign-owned resource-based or manufacturing firms rather

than on supporting domestic technology ventures. Innovation policy has typically emphasized multinational enterprises and relied on technology imported from the United States (Britton, 2003; Nicholson, 2016; Wolfe, Difrancesco, and Denney, 2022), overlooking the potential contributions of homegrown firms (Wolfe, 2015). More recently, however, scale-up and entrepreneurial firms have gained prominence in policy discourse, highlighted by federal investments in 2019 explicitly aimed at scaling up Canadian companies as part of a broader innovation agenda (ISED, 2019a).

This paper examines the landscape of Canada’s scale-up firms and the effectiveness of innovation policy in supporting their growth. We build on recent research (Denney, Vu, and Kelly, 2021) that mapped Canada’s high-performing firms and integrate new analysis of how these firms interact with government support programs. The goal is to provide a comprehensive, policy-oriented view of Canadian scale-ups – who they are, their economic contributions, and the extent to which current policy support aligns with their needs. Specifically, we ask: How prevalent are scale-ups in Canada, and in which industries and regions? What is their contribution to employment, output, and innovation compared to other firms? What characteristics (e.g., wages, research and development [R&D], exporting) distinguish scale-ups? And using the Business Innovation and Growth Support (BIGS) database at Statistics Canada, to what extent does Canada’s innovation policy mix (Denney, Southin, and Wolfe, 2023) – including both indirect Scientific Research and Experimental Development (SR&ED) tax credit and direct spending programs – support these high-growth firms?

Our analysis shows that while Canadian scale-ups are rare, they exert an outsized impact on jobs, exports, and innovation. Yet most never receive federal innovation support, and those that do are typically more mature, established firms rather than younger firms at earlier stages of scaling. SR&ED credits tend to reach firms before or during scaling, whereas BIGS funding generally arrives after scaling, reinforcing growth rather than catalyzing it. The impacts of support are positive. They raise employment, revenues, wages, SR&ED, and exports, but support is concentrated among a narrow subset of firms. These findings

highlight the need for a more cohesive and better-timed policy mix, one that balances early-stage enabling with growth-stage reinforcement, and that recognizes diverse scaling pathways beyond the technology sector.

The article proceeds as follows. Section 2 reviews the literature on scale-ups, high-growth firms, and innovation policy, drawing on both international evidence and the Canadian context. Section 3 outlines the data sources and methodology, including definitions of “scale-up” and the empirical strategy. Section 4 presents results in two parts: (1) a descriptive analysis of scale-ups’ economic footprint and characteristics, and (2) findings on the relationship between policy supports and firm scaling. Section 5 discusses implications for innovation policy and firm growth strategies in Canada. Section 6 concludes.

2 The Scale-Up Phenomenon and Innovation Policy

2.1 What is a scale-up?

How do we define scale-ups or high-growth firms? There is no single agreed-upon definition of “scale-up,” and various studies use different criteria to identify high-growth firms (HGFs). Broadly, scale-ups are firms that achieve rapid growth over a given period, but the specific growth metrics and thresholds vary. The OECD’s definition – widely adopted in research – classifies a high-growth enterprise as any firm with at least 10 employees at the beginning of a three-year period that achieves an average annualized growth in employees or revenue greater than 20 percent per year. This definition captures the magnitude and consistency of growth and includes a minimum initial size to avoid counting small base growth (Ahmad, 2006; Eurostat and OECD, 2008). By contrast, the Kauffman Foundation defines “growth entrepreneurship” in part by firms that reach certain revenue or employment size within a fixed time since startup (Morelix, Reedy, and Russell, 2016). For example, one Kauffman metric identifies firms that achieve at least \$2 million in revenue within 5 years of founding – emphasizing both the attainment of a growth threshold and the growth rate. Other studies

define HGFs as the top 5 percent or 10 percent of firms by growth in a given period (e.g., Clayton, Sadeghi, Talan, and Spletzer, 2013) or as those that exceed a certain absolute increase in employees (Birch, 1979; Schreyer, 2000).

Crucially, the choice of definition influences which firms are identified as scale-ups and the contributions observed. Daunfeldt, Elert, and Johansson (2014) demonstrate that revenue-based versus employment-based criteria can yield different conclusions about firm impact. Employment and revenue growth often diverge, highlighting three distinct properties of growth: trajectories (rate of growth), thresholds (absolute size), and consistency (duration). Recent work emphasizes the need for clarity in defining scale-ups, as different definitions identify distinct firm profiles (Coad et al., 2014; Peterson and Ahmad, 2007).

2.2 Characteristics and Economic Impact of Scale-Ups

Notwithstanding definitional nuances, a body of evidence demonstrates that high-growth firms, however defined, are disproportionately important for economic growth. Early work by Birch (1979) found that a small number of fast-growing “gazelles” generated the majority of net new jobs in the U.S., a finding echoed in later studies. In Canada, Picot and Dupuy (1998) reported that job creation is highly concentrated in a tiny fraction of firms, and more recent analysis confirms that firms growing employment rapidly make outsized contributions to overall employment levels. One Canadian study (Côté and Rosa, 2017) reported that, on average, employment-based scale-ups employ five to ten times as many people as comparable non-scale-ups. Similarly, high-growth firms contribute significantly to aggregate productivity gains and output growth (Haltiwanger et al., 2017). These firms often innovate and drive industry dynamism (Coad and Rao, 2008). Scale-ups commercialize new technologies, enter export markets, and spur upstream and downstream job creation (through supplier and consumer spending effects), amplifying their impact (Mason and Brown, 2013; Guzman and Stern, 2017).

Scale-ups are often described as “the guarantors of national prosperity”, driving innova-

tion and global competitiveness (Audretsch, 2012). The literature emphasizes that they are diverse—young or old, tech or non-tech, manufacturing or services—and that high growth may or may not persist. What unites them is exceptional performance within a given time frame.

Several factors are associated with a firm’s likelihood of scaling up. One is innovation capability – firms that innovate (develop new products, processes, or organizational or business model innovations) are more likely to achieve high growth (Coad and Rao, 2008; Stam and Wennberg, 2009). Engaging in R&D and holding patents are often positively correlated with subsequent growth, especially in knowledge-intensive sectors (Hölzl, 2009; Demir, Wennberg, and McKelvie, 2017). Another is export market penetration: exporting SMEs tend to grow faster as they tap into larger markets (Love and Roper, 2015). Human capital and skills also play a role; high-growth firms are led by ambitious, experienced management teams and employ skilled workers, enabling them to exploit growth opportunities (Barringer et al., 2005). Additionally, external factors such as industrial clusters and networks can support scaling – for instance, being located in a dynamic region with rich knowledge spillovers can facilitate firm growth (Spencer et al. 2010; Bramwell and Wolfe, 2016; Denney, Southin and Wolfe 2021). However, it is important to note that high growth can occur in any industry sector; Canadian data show that scale-ups are not limited to tech industries: by one recent estimate, only about 8–9 percent of Canadian scale-ups are in the information and communications technology (ICT) sector. The majority are found in other sectors, ranging from manufacturing to services. This distribution suggests that a national scale-up strategy should have a broad sectoral focus.

2.3 Scale-Ups and Policy Support

The substantial economic contribution of scale-ups has led policymakers to endorse the need for more firms to scale and to support those that do. If scale-ups drive disproportionate gains in jobs, productivity, and innovation, fostering additional HGFs could yield consider-

able economic benefits (Henrekson and Johansson, 2010). This rationale has led to targeted support programs for high-growth firms in many countries (OECD, 2013). Approaches vary: Some policies aim to improve the general business environment (e.g., reducing financing constraints, regulatory burdens, or skill shortages), while others target scale-ups with tailored interventions (Mason and Brown, 2013). Examples of targeted support include mentorship and accelerator programs for rapidly growing firms, government venture funds or growth capital programs, and scale-up advisory services. The UK’s ScaleUp Institute was established to research and promote scale-up-friendly policies (ScaleUp Institute, 2019), while the Horizon Europe program includes instruments to support “innovation champions” poised for high growth.

Canada’s innovation policy has historically relied on indirect support for business R&D (notably the Scientific Research and Experimental Development (SR&ED) tax credit) and broad programs for SMEs, rather than targeted scale-up programs (Bérubé and Therrien, 2016). Indirect incentives like tax credits benefit many firms undertaking R&D and tend to favor established companies with taxable income. In recent years, however, Canadian policy has focused more on direct support and targeted initiatives for high-growth firms. The Industrial Research Assistance Program (IRAP) has long provided grants and technical advice to innovative SMEs, including some that become scale-ups. The government’s 2019 announcement of new investments to “scale up Canadian companies” included funding for incubators and financing programs aimed at firms poised for growth. There are also venture funds (e.g., the Venture Capital Catalyst Initiative and various provincial funds) to help firms grow. While Canada’s policy mix still favours indirect support (tax credits), there has been a gradual shift to direct interventions to support scaling firms (Innovation, Science and Economic Development [ISED], 2019b).

A key question is how well these supports align with the needs and behaviour of scale-up firms. Recent research by Denney, Southin, and Wolfe (2021) suggests that Canadian scale-up entrepreneurs perceive gaps in the policy mix – notably, a need for more targeted

support and growth-stage financing instead of generic SME measures. Founders of high-growth firms emphasize the importance of risk capital (e.g., venture funding), access to talent, expansion into global markets, and procurement support, areas where policy can play a facilitating role (Clemens et al., 2015; Council of Canadian Innovators, 2021). Evidence from Dowsett and Kelly (2025) further underscores that firms with stronger pre-support performance and innovation capacity tend to achieve more pronounced outcomes from public programs, suggesting that higher-performing firms are better able to translate funding into growth. At the same time, some scholars caution that governments attempting to “pick winners” might misallocate resources or introduce distortions (Nightingale and Coad, 2014). The challenge for public policy is to support potential scale-ups in an evidence-based way that complements market mechanisms, rather than chasing the latest high-profile tech unicorns.

In sum, the literature provides several insights to guide our analysis. First, scale-ups can be defined in multiple ways, and research should consider both employment and revenue growth perspectives. Second, while rare, scale-ups have disproportionate economic impacts – contributing significantly to job creation, productivity, and innovation – which justifies policy attention. Third, scale-ups are heterogeneous and span sectors; policies should be broad enough to include high-growth firms in both traditional and high-tech industries. Fourth, innovation (R&D, patents), exporting, and strong management emerge as common enablers of scaling up, suggesting these may be levers for policy (e.g., programs that enhance firm innovation capacity or global reach could help more firms grow). Finally, the role of policy support for scale-ups is nuanced: many high-growth firms may reach scale-up status without direct government aid, yet well-designed support (targeted funding, services) can reduce barriers that impede further growth (such as financing gaps for young firms or the high risk of R&D investments). The article draws on these insights to examine which firms scale in Canada and how the current innovation policy mix supports them.

3 Data and Methods

This study employs a comprehensive econometric framework using administrative data provided by Statistics Canada, covering firm-level information from 2007 to 2019. The dataset integrates multiple sources, including Statistics Canada’s Business Register, administrative tax records, and records of federal innovation support programs – specifically, the Business Innovation and Growth Support (BIGS) database and the Scientific Research and Experimental Development (SR&ED) tax credit database. BIGS captures direct government support mechanisms such as grants and contributions from programs, including the Industrial Research Assistance Program (IRAP) and regional development agencies. SR&ED data reflect indirect support for innovation through tax incentives. Additionally, patent data and R&D expenditure records are utilized to categorize firms engaging in innovation-related activities. All data linkages and analyses were securely conducted within Statistics Canada’s research data centers, ensuring confidentiality and anonymization.

The methodological approach comprises two primary analytical components: (1) descriptive analysis of scale-up firm characteristics and contributions, and (2) econometric analysis evaluating the impact of government innovation support on scale-up performance.

For the descriptive analysis, we identify all scale-up firms in the data (2010–2019) under each definition and examine their prevalence, economic footprint, and distinguishing characteristics. Specifically, we calculate the share of all firms classified as scale-ups, their contribution to total private-sector employment and revenue, and their distribution across industries and regions. We then compare scale-ups to otherwise similar non-scale-up firms in terms of average size, age, wage levels, and productivity (measured as revenue per employee). Finally, we assess the proportion of scale-ups that export, perform R&D, or hold patents relative to the broader firm population to highlight the traits most associated with rapid growth.

For descriptive profiling, we identify scale-up firms using two primary definitions: the OECD criterion, which includes firms with at least ten employees that achieve a minimum

of 20 percent annualized employment growth over three consecutive years, and a revenue-based definition inspired by the Kauffman Foundation, which includes firms achieving a minimum of 20 percent annualized revenue growth over three consecutive years and exceeding CAD 2 million in annual revenue. Scale-up prevalence, industry and regional distribution, and economic impacts – including employment, revenue contributions, wage levels, labour productivity, R&D intensity, patenting activity, and export orientation – are systematically compared to the broader firm population to illustrate distinct characteristics and contributions of scale-up firms. Table 1 summarizes the definitions used and the populations they include.

Growth Dimension	Shorthand	Firm population	Definition
Employment	OECD Employment	Established firms	Average of at least 20 percent year-on-year growth in employment for three consecutive years At least 10 employees at the beginning of the growth period
Revenue	Kauffman Revenue	All firms	$\geq$ 20 percent year-on-year growth in revenue over three years $\geq$ \$2 million in revenue at the end of the measurement year

Table 1: Scale-up Definitions

Note: Table summarizes the two scale-up definitions used in the analysis. Thresholds are expressed in percent terms and reflect firm-level administrative data.

The econometric analysis employs a quasi-experimental design using propensity score matching combined with firm-level fixed-effects regression models. We focus on the impact of receiving direct government support (BIGS) post-initial scaling, comparing supported scale-ups to similar firms that also scaled previously, but either received no support or received support only after our selected five-year performance measurement window for their matched treated firm.

3.1 Step 1 — Propensity Score Matching (PSM)

Define $BIGS_i = 1$ for firms that receive BIGS for the first time in our focal cohort (2012–2014) and $BIGS_i = 0$ otherwise. We estimate each firm’s propensity score using a logistic regression based on pre-treatment characteristics:

$$\Pr(BIGS_i = 1 \mid X_{i,t-1}) = \Lambda(\alpha + \alpha_t + \alpha_{\text{ind}} + \alpha_{\text{reg}} + \beta' X_{i,t-1}), \quad (i, t) \in P, \quad (1)$$

where α_t , α_{ind} , and α_{reg} denote year, industry, and region fixed effects. The vector $X_{i,t-1}$ includes pre-treatment firm size, wages, R&D spending, export activity, years since scaling, and pre-support growth trajectories. Following standard practice, the propensity score is estimated on a panel including all control observations, while treated firms appear only in the pre-treatment years up to their first BIGS award. Formally:

$$P = \{(i, t) : i \in T, t \leq T_i^{\text{treat}}\} \cup \{(i, t) : i \in C, \text{ all } t\}. \quad (2)$$

Matching is implemented via nearest neighbour within a caliper: for each treated firm, we select an untreated firm with a similar *linearized* estimated propensity score, provided it lies within a pre-specified caliper of 0.2 standard deviations of the treated distribution. Matching is conducted with replacement, and treated firms without an acceptable match are dropped. Potential controls are further restricted to the same three-digit industry and region as the treated firm. Balance is assessed using standardized differences and distributional tests.

This matching strategy constructs a credible counterfactual by identifying comparable untreated firms, thereby reducing selection bias and enhancing internal validity. Restricting the analysis to firms that have already achieved scale-up status mitigates concerns about selection on managerial capability or resourcefulness. Within this framework, all firms have reached comparable growth milestones, suggesting similar underlying capabilities.

The resulting set of treated firms and their matched controls forms the matched sample M , which serves as the basis for the fixed-effects analysis that follows.

3.2 Step 2 — Fixed-effects estimation on the matched sample

Using the matched sample M , we estimate a fixed-effects model over a ten-year window, covering the five years before and the five years after the match year T_i^* (the year firms receive BIGS support). The estimating equation is:

$$Y_{it} = \alpha + \beta \text{Support}_{it} + X'_{it}\gamma + \mu_i + \theta_{\text{ind} \times t} + \varepsilon_{it}, \quad (i, t) \in M, \quad |t - T_i^*| \leq 5. \quad (3)$$

Here, Y_{it} denotes the outcome variable (employment growth, revenue growth, R&D intensity, export activity) for firm i at time t . The term μ_i represents firm-specific fixed effects, capturing unobserved, time-invariant firm characteristics. The term $\theta_{\text{ind} \times t}$ represents industry-year interactions capturing sector-specific macroeconomic shocks. Support_{it} is a binary indicator equal to one if firm i receives direct government support at time t (post-initial scaling) and remains one thereafter. The vector X_{it} includes time-varying controls such as patent applications and SR&ED tax credit support. The disturbance term ε_{it} is the idiosyncratic error.

The primary parameter of interest, β , measures the average treatment effect of government support on firm outcomes. Robust standard errors are clustered at the firm level to address serial correlation and heteroskedasticity.

The empirical strategy mitigates endogeneity and selection bias through matched controls and fixed-effects estimation, reducing biases from observable and time-invariant unobservable factors. Nevertheless, some caution is warranted: unobserved time-varying factors may still influence both support receipt and outcomes. Results are reported using conventional significance thresholds (five percent, with robustness at one and ten percent). Sensitivity checks using the staggered difference-in-differences estimator of Callaway and Sant’Anna (2021) yield qualitatively and quantitatively similar findings.

The analysis is grounded in Penrose’s (1959) theory of firm growth, which emphasizes

the role of internal capabilities (innovation, managerial skill, productivity) and external resources (finance, market opportunities). Government innovation support may alleviate growth constraints—such as financing gaps or insufficient R&D capacity—and thereby facilitate accelerated or sustained growth. Conversely, selection effects, whereby governments tend to support firms already on strong growth trajectories, are explicitly accounted for in the empirical design.

Overall, this combined methodological approach provides a rigorous evaluation of the role played by government innovation support in fostering and sustaining scale-up growth, contributing evidence to policy debates and the broader innovation economics literature.

4 Findings

4.1 Profile of Scale-Up Firms in Canada

We profile Canadian scale ups over 2010–2019 using two complementary definitions introduced in the previous section: (i) OECD employment and (ii) a revenue based threshold. These capture, respectively, labour-absorbing vs. revenue /productivity-intensive scaling trajectories.

Scale-ups constitute a small fraction of the business landscape, yet their economic impact is disproportionately large. Regardless of definition, only a limited share of eligible firms exhibit the rapid growth required to qualify as scale-ups, consistent with longstanding evidence on the “vital few” phenomenon (Birch et al., 1995; NESTA, 2009). As Figure 1 shows, incidence rates remain low when benchmarked against each definition’s relevant population: roughly 6–7 percent of OECD-eligible firms meet the employment-based scale-up criterion in a given year, while only about 1.5–2 percent of firms satisfy the Kauffman revenue-based definition. These figures underscore that scale-ups are rare even within their own definitional pools, implying that policies aiming to increase their numbers must contend with inherently low base rates.

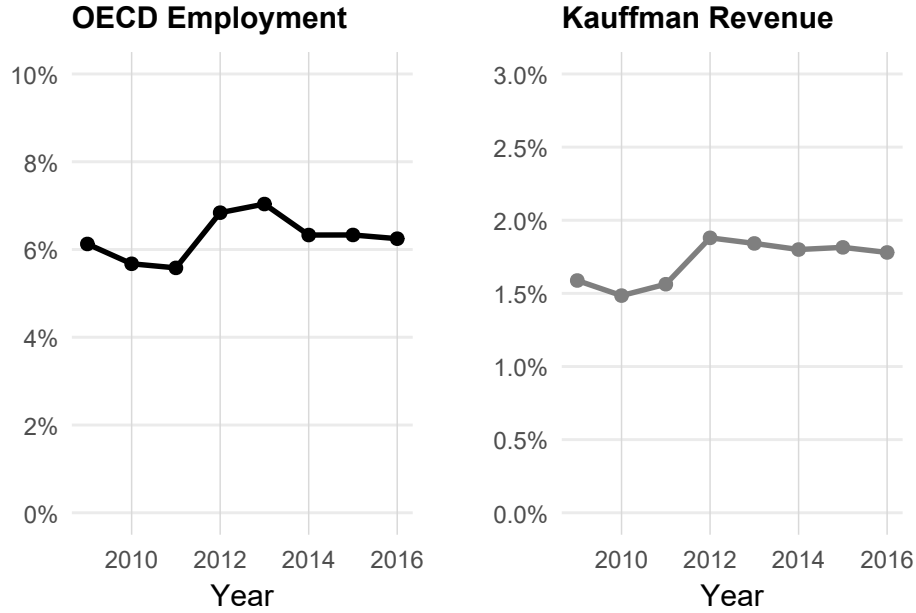


Figure 1: Scale-up incidence within definition-specific firm populations, 2009–2019

Note: Each panel reports annual scale-up rates using a different definition: the OECD Employment definition (left) and the Kauffman Revenue definition (right). Incidence is calculated as the share of firms *eligible under that definition* that qualify as scale-ups in a given year. Because the two definitions apply to different underlying firm populations, the levels are not directly comparable across panels.

Despite their limited prevalence, Canadian scale-ups account for a disproportionately large share of economic activity. International evidence shows that fewer than five percent of firms typically generate a substantial share of net new jobs and output growth (Birch et al., 1995; NESTA, 2009; Anyadike-Danes et al., 2015), and Canada is no exception. The incidence patterns reported in Figure 1 align with this broader empirical regularity: only a small subset of firms—whether defined by employment growth or revenue expansion—achieves scale-up status, yet these firms play an outsized role in driving aggregate performance. The implication is that even modest changes in the number of scale-ups can meaningfully influence economic outcomes.

Yet these firms exert an outsized impact on employment and revenue. Figure 2 demonstrates that while scale-ups account for only a sliver of the overall firm population, they contribute a substantially larger share of total private-sector employment. Employment-

based scale-ups employ, on average, five to ten times as many workers as otherwise similar non-scaling firms, and revenue-based scale-ups similarly outperform in terms of revenue margins. Scale-ups' contribution to total revenue is even more striking: average revenue per firm among Kauffman-revenue scale-ups is several multiples higher than among non-scale-ups, consistent with prior evidence that the fastest-growing firms generate an outsized share of incremental sales. In this sense, the data here reinforces the central point that scale-ups, though rare, are critical drivers of Canada's economic dynamism.

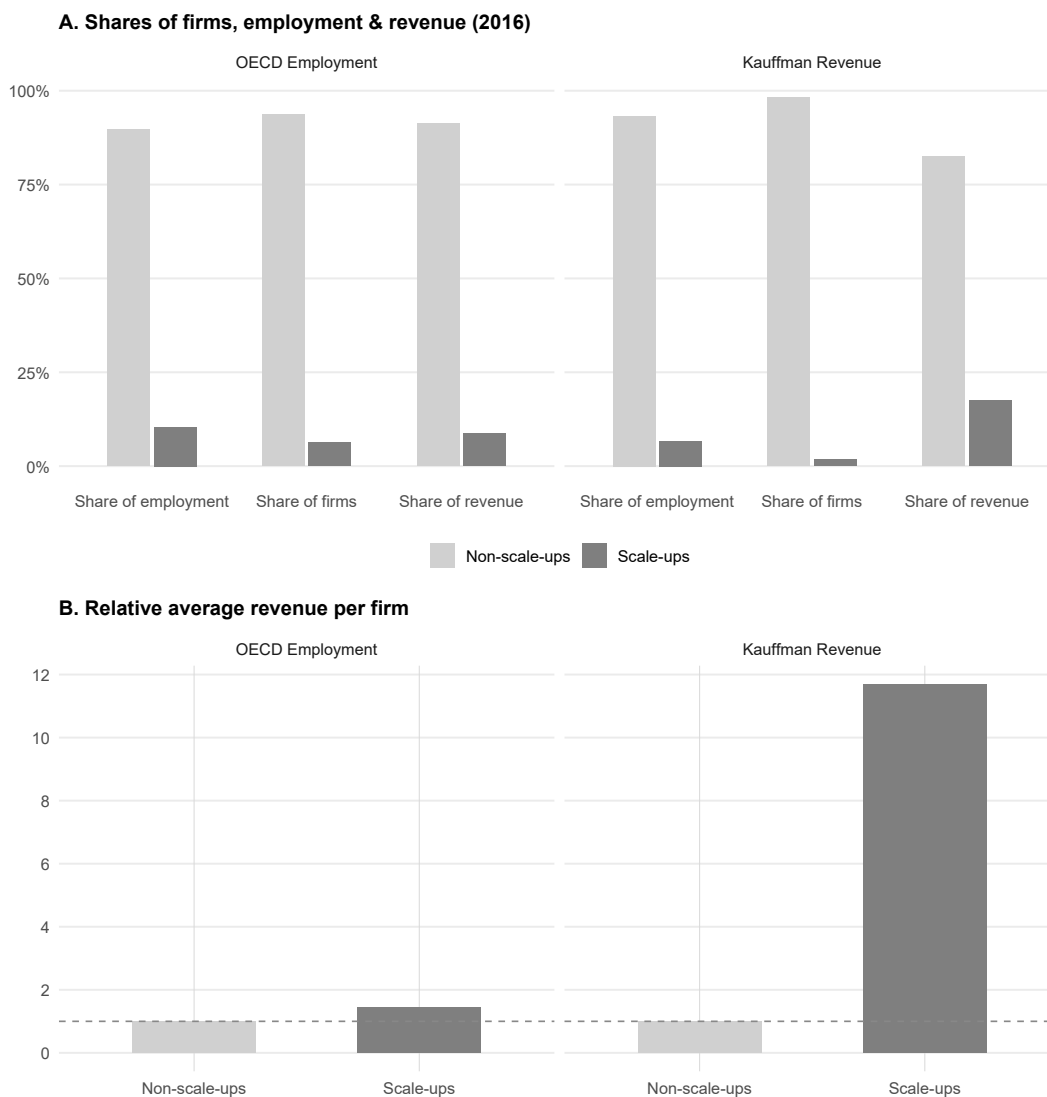


Figure 2: Economic footprint of scale-ups

Note: Panel A shows the share of firms, employment, and revenue accounted for by scale-ups and non-scale-ups in 2016, under both the Kauffman Revenue and OECD Employment definitions. Panel B reports relative average revenue per firm, where values are normalized to non-scale-ups (= 1), so numbers reflect how many times larger the average revenue of scale-ups is. Panel A values are expressed in percent terms; Panel B values are index values.

Scaling is not confined to the technology sector. Although much of the policy discourse tends to equate scale-up activity with technology-based firms (Nightingale and Coad, 2014; Research Money, 2024), only about 8–9 percent of Canadian scale-ups operate in ICT or related fields; the largest shares are in services (professional services, wholesale/retail), construction, and selected areas of manufacturing. Figure 3 illustrates this industry mix. In policy terms, high growth is not exclusive to “tech champions”: scale-ups are found in food services, construction, and traditional manufacturing alongside software and advanced industries (Henrekson and Johansson, 2010). Sectoral differences also shape how firms scale: hospitality and retail scale-ups tend to be employment-rich but less productive, while tech and finance scale-ups achieve revenue and productivity growth with fewer new jobs. This trade-off underscores that scale-ups come in different “types”, such as employment-intensive versus productivity-intensive, and no single type maximizes all objectives simultaneously.

Geographically, scale-ups cluster in major urban regions such as Toronto–Waterloo, Montreal, Vancouver, Ottawa, and Calgary. When normalized by the number of firms, smaller centres like Kitchener–Waterloo and Ottawa stand out, reflecting strong ecosystems and cluster spillovers (McKinsey 2016; Creutzberg, Ornston and Wolfe 2024; Spencer et al. 2010). These patterns suggest that clusters amplify scaling, but nationwide programs remain important so that promising firms outside large hubs can also access support.

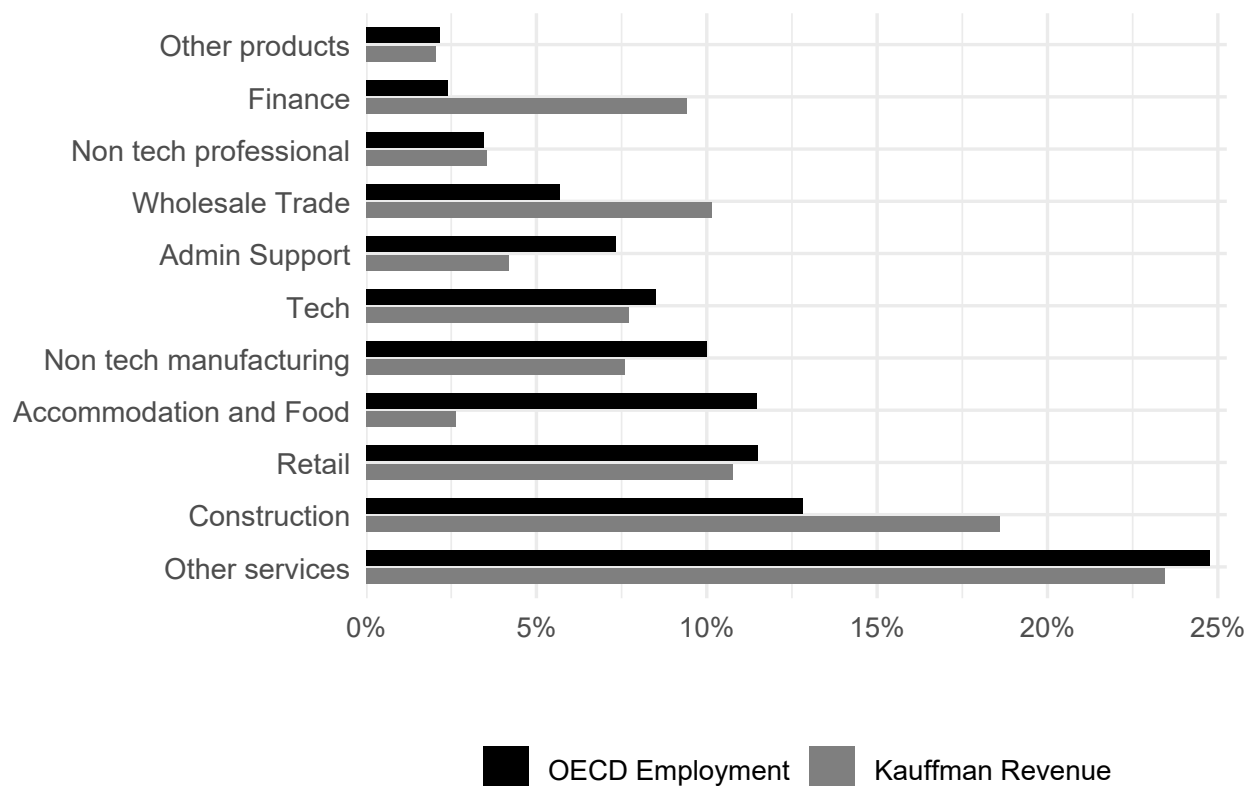


Figure 3: Distribution of scale-ups across industries)

Note: Figure shows the distribution of all scale-ups across industries in 2016 under two definitions: the OECD Employment definition and the Kauffman Revenue definition. Values represent the percentage of all scale-up firms in each industry.

On wages and productivity, the evidence is mixed. In aggregate, scale-ups tend to pay somewhat higher wages, but gaps vary by sector and definition. Tech scale-ups consistently paid above average, whereas low-margin service-sector scale-ups did not. Productivity (measured as revenue per worker or average productivity growth) is generally higher among revenue-based scale-ups, though employment-intensive scale-ups often lag due to rapid hiring. Firm-level productivity growth is nevertheless strong during scaling, reflecting efficiency gains or economies of scale (Table 2).

Scale-ups are also more innovative and internationally oriented than other firms. About 30 percent report R&D or patents (versus approx. 5 percent of non-scalers), and more than half are exporters (versus 10–15 percent of all firms). These features are strongly correlated

with faster growth and greater resilience, though not prerequisites: some construction or retail firms scale without formal R&D or IP. Still, innovation and exporting are common features that distinguish many scale-ups (Table 3).

Average Pay Differentials	Non-Scale-Ups	Scale-Ups
OECD Employment	\$52,400	\$49,600
Kauffman Revenue	\$55,600	\$64,000
Average Productivity Growth	Non-Scale-Ups	Scale-Ups
OECD Employment	−1.67 <i>percent</i>	2.17 <i>percent</i>
Kauffman Revenue	−2.69 <i>percent</i>	8.78 <i>percent</i>

Table 2: Wages and Productivity, Scale-Ups vs. Non-Scale-Ups (2016)

Note: Table reports average pay and average productivity growth (APG) for scale-ups and non-scale-ups under the OECD Employment and Kauffman Revenue definitions. APG values represent percent changes.

Share of Exporters	Non-Scale-Ups	Scale-Ups
OECD Employment	14.60 percent	18.37 percent
Kauffman Revenue	2.62 percent	17.28 percent
Share of R&D Spenders	Non-Scale-Ups	Scale-Ups
OECD Employment	6.18 percent	11.55 percent
Kauffman Revenue	1.14 percent	8.01 percent

Table 3: Innovation and Internationalization, Scale-Ups vs. Non-Scale-Ups (2016)

Note: Table reports exporter shares and R&D participation rates for firms classified as scale-ups and non-scale-ups under the OECD Employment and Kauffman Revenue definitions. Shares represent the percent of firms in each group engaging in exporting or R&D activities.

In sum, Canadian scale-ups constitute a small yet pivotal share of the firm population, making a disproportionate contribution to employment and output growth. They are distributed across diverse sectors and regions, often innovate and export, and follow divergent growth paths. Some are driven primarily by employment expansion, others by productivity and innovation gains. This heterogeneity underscores that no single model of scaling prevails, with direct implications for the design and assessment of innovation policy.

Against this backdrop, we now turn to examine how scale-ups interact with government support mechanisms. Building on this descriptive profile, the following section shifts from what scale-ups look like to how they interact with Canada’s innovation policy system. We first describe which firms receive federal innovation support and at what stage of their growth, drawing on administrative data linking the Business Innovation and Growth Supports (BIGS) and SR&ED tax credit programs. We then assess the impact of this support using an econometric framework designed to isolate the effects of government assistance on subsequent firm performance.

4.2 Interaction with Innovation Policy Supports: Who Gets Help and When?

This subsection examines how Canadian scale-ups interact with federal innovation supports. We link Statistics Canada firm-level performance data to program records from Business Innovation and Growth Supports (BIGS) and the Scientific Research and Experimental Development (SR&ED) tax credit. This allows us to assess which firms obtain support, at what stage of their growth trajectories they receive it, how much support they capture, and what impacts can be detected on subsequent outcomes.

A central policy question is whether federal innovation programs effectively reach scale-ups, and whether support arrives when it can matter most. Linking Statistics Canada administrative data with records from Business Innovation and Growth Supports (BIGS) and SR&ED shows the coverage, timing, intensity, and impacts of support on firms that scale.

In terms of coverage, most scale-ups do not receive federal innovation support. In 2019, only 7–11 percent had accessed BIGS or SR&ED; when following the 2012–2015 cohort over 2007–2019, cumulative participation rises to 15–23 percent (Figure 4). Thus, roughly three-quarters of scale-ups grew without direct federal assistance, which is important context for interpreting impacts.

Support is also not evenly distributed across scale-ups. Uptake is much higher among firms with innovation markers, particularly among larger innovators. Between 72-77 percent of patenting scale-ups and 54-57 percent of R&D performers accessed BIGS at some point (Figure 5). Among large innovative firms (100+ employees) that have scaled, uptake is even higher, reaching 85-86 percent and 69-70 percent, respectively. These patterns indicate that direct support disproportionately flows to firms that are already innovation-intensive and further along the growth path. A notable minority of BIGS recipients had already scaled before first receipt (approx. 20–30 percent, program/definition-specific), which is consistent with programs that emphasize demonstrated performance (Appendix Figure A.1).

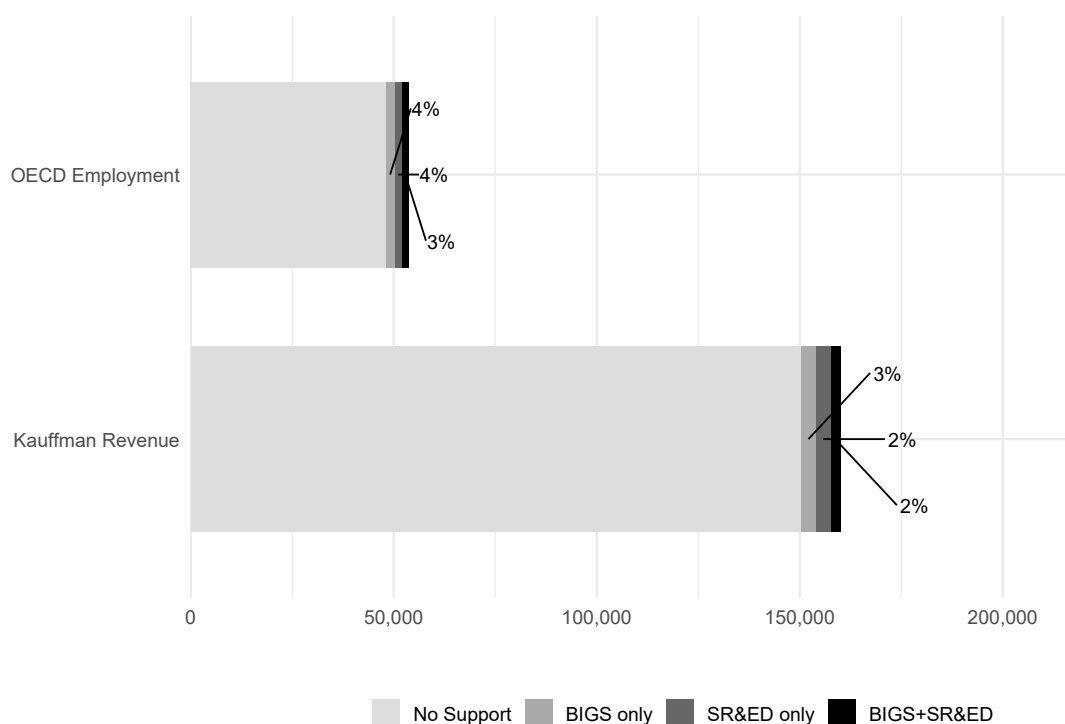


Figure 4: Support status of scale-ups in 2019

Note: Figure shows the distribution of support pathways among scale-ups in 2019, based on the 2010–2019 firm cohort. Counts are shown separately for scale-ups defined under the Kauffman Revenue definition and the OECD Employment definition. “No Support” represents firms that received neither BIGS nor SR&ED; “BIGS only” and “SR&ED only” represent firms receiving support exclusively through those programs; and “BIGS+SR&ED” represents firms receiving both. The figure highlights how most scale-ups grow without direct program support, while only a small share receive targeted innovation subsidies.

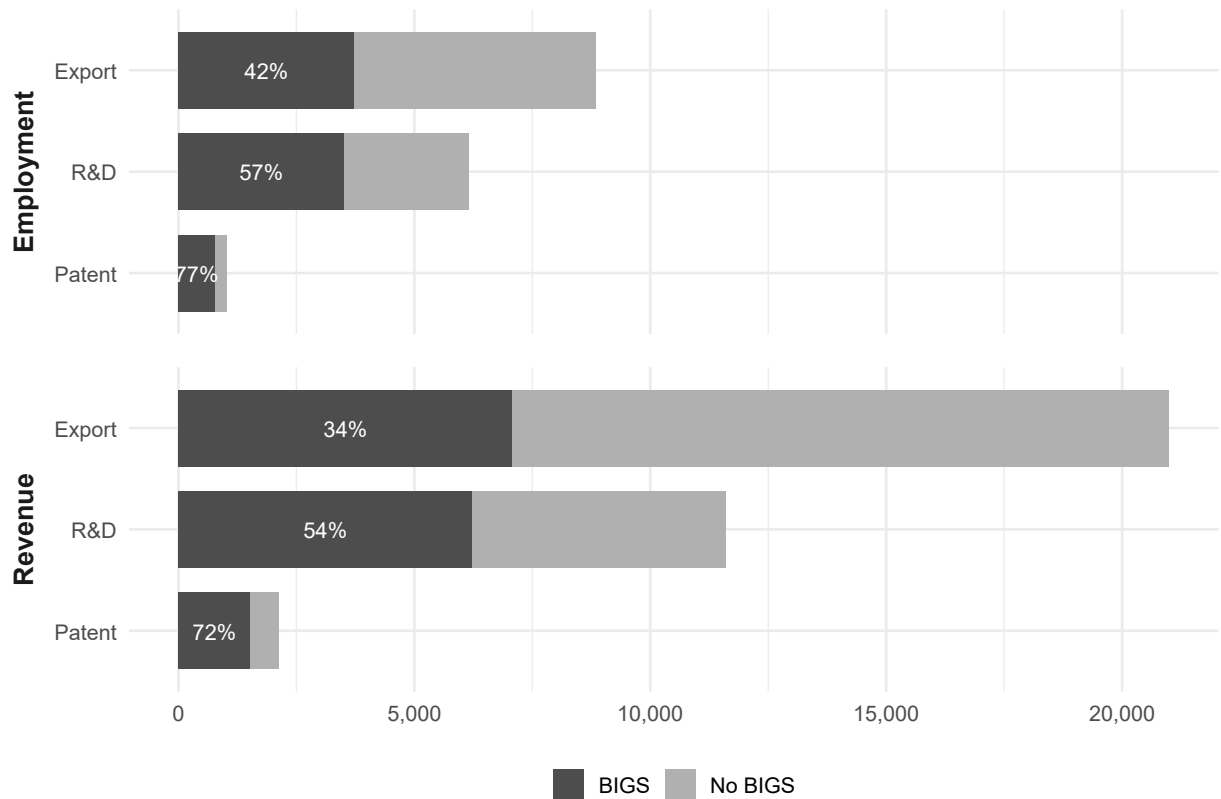


Figure 5: BIGS participation among innovative scale-ups (2010–2015 cohort)

Note: Figure reports counts of scale-ups that engaged in exporting, R&D, or patenting and indicates whether they received support through BIGS at any point during 2007–2019. A firm is classified as an exporter or R&D performer if it recorded exports or R&D expenditures in any year during 2007–2019, and as a patentee if it submitted at least one patent application during 2007–2015. The figure highlights the prevalence of BIGS participation among innovative scale-ups and contrasts supported and unsupported firms within each innovation category.

The timing of support also differs across instruments. More than half of BIGS-supported scale-ups first received funding after their initial scaling episode, while about 86 percent of SR&ED-supported scale-ups claimed credits before or during scaling (Figure 6). This divergence reflects the different mandates and structures of the two programs. SR&ED, as a tax credit tied to qualifying R&D expenditures, is accessible to firms with ongoing research activities and therefore tends to precede or coincide with scaling. BIGS programs, by contrast, rely on application and adjudication processes that often privilege firms with demonstrated capacity or growth trajectories, meaning that funding typically arrives once a firm is already expanding.

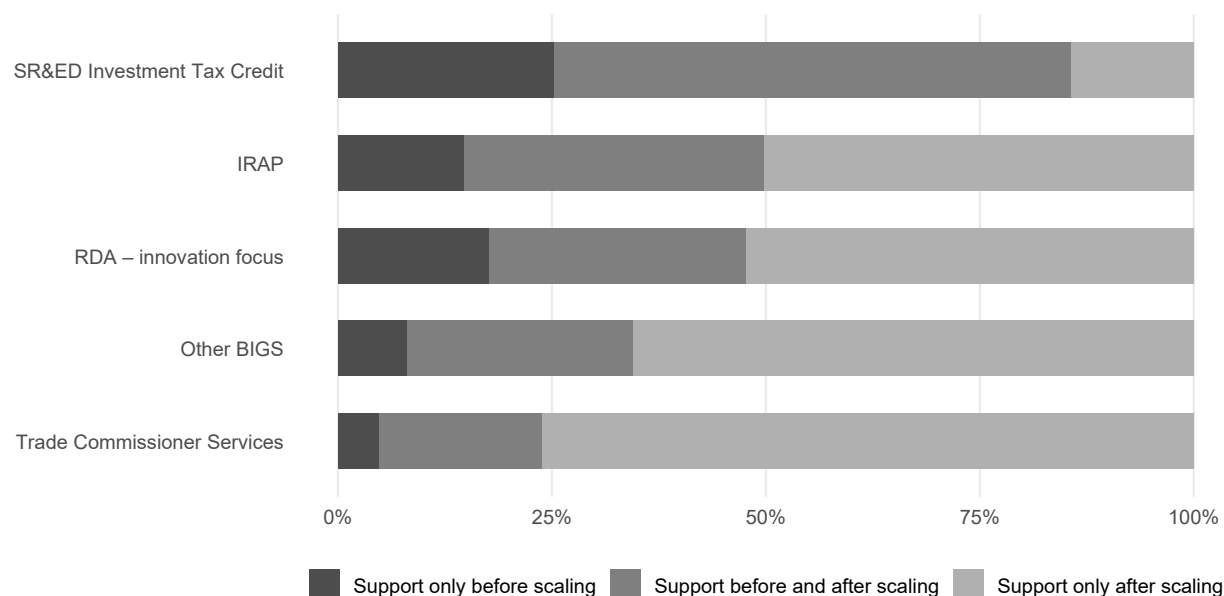


Figure 6: Timing of BIGS and SR&ED support relative to initial scaling (2012–2015 cohort, OECD employment definition)

Note: Figure shows the distribution of scale-ups by the timing of first receipt of BIGS and SR&ED support relative to the initial scaling episode. Firms are categorized according to whether support was received only before scaling, both before and after scaling, or only after scaling. The analysis is restricted to scale-ups in the 2012–2015 cohort, classified using the OECD employment definition.

Despite these differences in first-touch, the bulk of dollars for both instruments flows to firms after they have scaled (Figure 7). This partly reflects the mechanics of program design: larger firms with higher eligible expenditures qualify for larger SR&ED claims, and BIGS programs channel more funding to more established firms. It also reflects firm selection: many companies intensify their R&D and financing efforts once growth is underway, which triggers eligibility for higher-value support. The consequence is that federal innovation policy plays more of a reinforcing role than an initiating one, amplifying trajectories of firms already on a growth path rather than catalyzing scaling from the outset. From a policy perspective, this raises the question of whether the timing of support optimally aligns with firms’ needs during the high-risk scaling phase.

Timing also varies with age: mature scalers, defined as those that first scaled at 10 years or older, receive a majority of their BIGS value after scaling (Appendix Figure A.2). The

overall pattern is consistent with SR&ED acting as an early-stage enabler of R&D, and BIGS as a post-scaling reinforcement.

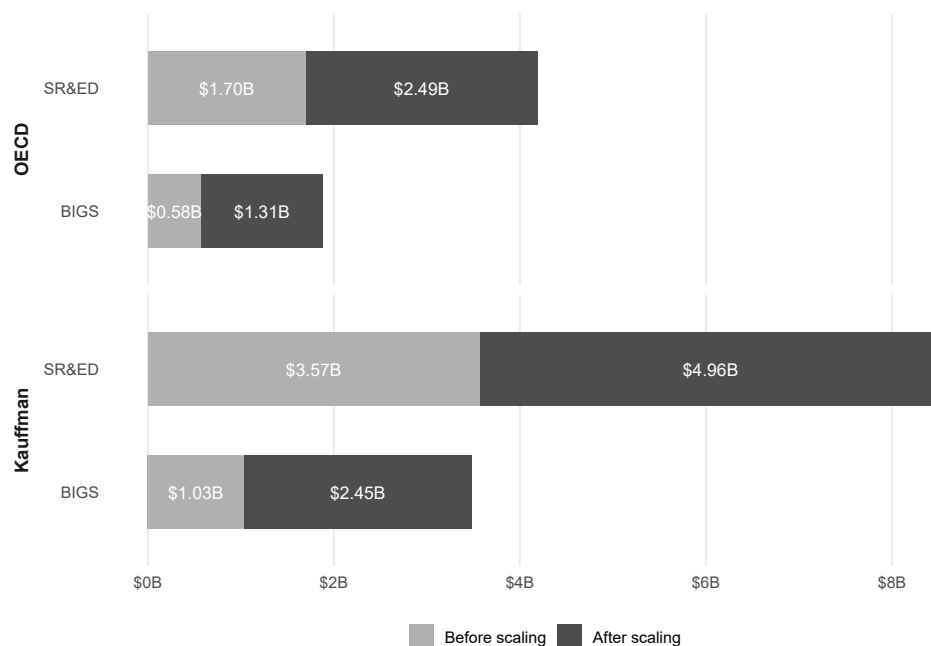


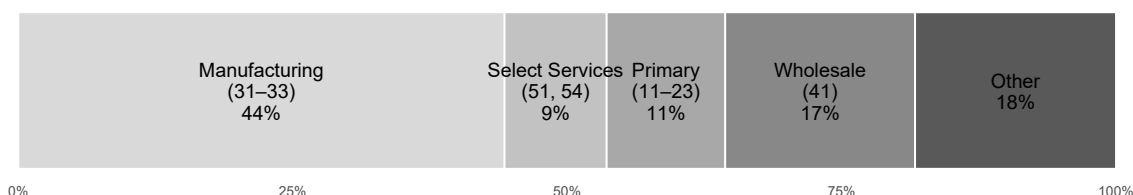
Figure 7: Total value of support (2007–2019) by timing relative to initial scaling, 2012–2015 cohort

Note: Figure reports the total value of BIGS and SR&ED support received by scale-ups in the 2012–2015 cohort, classified using the OECD employment definition. Support values reflect cumulative program receipts between 2007 and 2019. Firms are grouped by whether they received support before the initial scaling episode or only after scaling. BIGS and SR&ED values are shown separately for revenue-defined and employment-defined scale-ups.

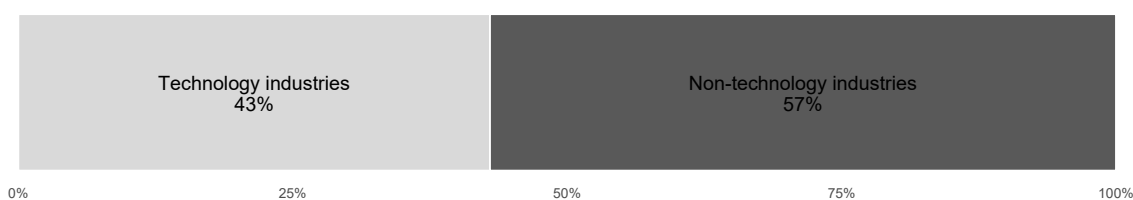
Support intensity and concentration also reveal distinct patterns. Continuous users that access programs both before and after scaling capture roughly three to four times the total support of one-stage users, indicating cumulative advantage. Allocation is concentrated by sector and size: about 70 percent of BIGS value to scale-ups goes to manufacturing, information and cultural industries, and professional, scientific, and technical services; around 60 percent goes to firms with more than 50 employees. Technology-intensive industries receive, on average, roughly twice as much BIGS funding per recipient firm as non-tech sectors, reflecting both higher award values and a greater concentration of support among these industries (Figure 8). Program mix matters as well: IRAP and Trade Commissioner Service

(TCS) are the most common touchpoints among scale-ups, although with smaller average award sizes, whereas programs such as Strategic Innovation Fund (SIF) and Sustainable Development Technologies Canada (SDTC) are less common but provide much higher value (Appendix Table A.1).

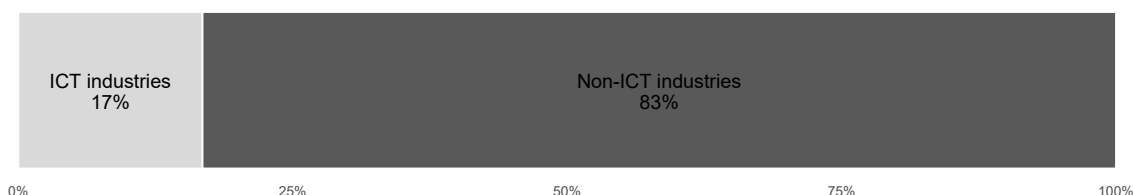
Industry sectors



Technology vs Non-technology



ICT vs Non-ICT



Firm size classes

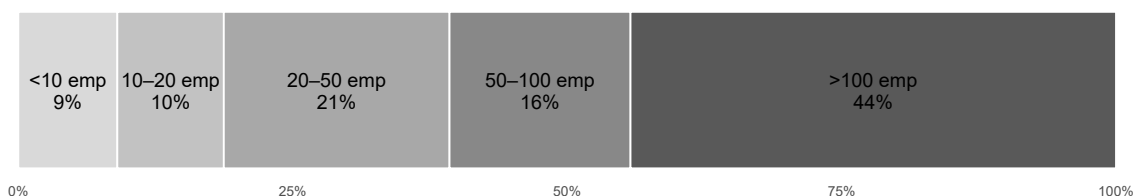


Figure 8: BIGS support value distribution for 2019 scale-ups (2010–2019 cohort, Kauffman revenue definition)

Note: Figure displays the distribution of BIGS support value across industry sectors, technology intensity, ICT status, and firm size for scale-ups in 2019 identified under the Kauffman revenue definition. The technology industry classification follows the set of 4-digit NAICS codes with at least three times the national average concentration of technology workers: 2211, 3333, 3341–3346, 3353, 3364, 4173, 4861–4862, 5112, 5122, 5152, 5171–5172, 5174, 5179, 5182, 5211, 5261, 5413, 5415, 5417, and 8112. Industry values represent the share of total BIGS support allocated to each group.

Having documented descriptive patterns of program uptake among scalers, we now turn to econometric analysis to estimate the treatment effects of BIGS support. We employ a propensity-score matching approach to identify comparable untreated firms and then estimate firm-level fixed-effects regressions on the matched sample. Results indicate that BIGS-supported firms exhibit 5–10 percentage-point higher annual employment and revenue growth over five years post-treatment relative to matched controls, and are more likely to undergo subsequent scaling episodes (Figure 9). We also find positive effects on average wages, R&D expenditure, and export intensity, though no near-term, statistically significant effect on labour productivity is observed, which is consistent with transitional “growing pains.” Program-level estimates show broadly consistent gains across BIGS instruments, suggesting similar marginal returns per dollar despite differing objectives and target populations (Appendix Table A.2).

Outcome	OECD employment		Kauffman revenue	
	ATT	Sig.	ATT	Sig.
Employment	10%	***	14%	***
Revenue	17%	***	20%	***
Scaling incidence	5%	***	4%	***
Average wages	2%	*	3%	***
R&D expenditures	15%	***	14%	***
R&D incidence	5%	***	5%	***
Exports	17%	**	12%	**
Export incidence	4%	***	5%	***
Sales per employee	2%		1%	

Table 4: ATTs of BIGS on Scale-Up Performance (Matched 2012–2014 Cohort)

Note: Table reports fixed-effects estimates of the aggregate treatment effect of BIGS for scale-ups under the OECD employment and Kauffman revenue definitions. Models include controls for R&D performers, exporters, patent holders, industry–year interactions, and SR&ED support. Treatment and matched control observations are included from $t - 5$ to $t + 5$. Standard errors are clustered at the firm level. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Finally, robustness checks give confidence in these estimates. Matching improves covariate balance (K–S tests) and visual inspection confirms pre-trends are flat (Appendix Table A.3);

estimates are robust to alternative calipers, matching ratios, and fixed-effects specifications incorporating additional controls for export and R&D activities. Many scale-ups also participate in multiple programs; we control for SR&ED and program aggregates, but spillovers across instruments may attenuate program-specific estimates. The measurement scope also remains a limitation, as service exports and procurement effects are under-captured; including BDC and EDC data would strengthen inferences on export-related effects.

In summary, most Canadian scale-ups never access federal innovation programs, and those that do are typically larger, innovation-oriented firms. SR&ED credits play a pre- and during-scaling role, while BIGS funding generally arrives after scaling. Support is concentrated by sector, firm size, and technology intensity, and firms that receive both early and later support accumulate much larger total benefits. Econometric evidence indicates that BIGS support contributes positively to employment and revenue growth, as well as to wages, R&D, and exporting, though without immediate productivity effects. These findings underscore that federal programs amplify the trajectories of a select subset of scale-ups, and that the timing and targeting of support carry significant policy implications.

5 Conclusion and Discussion

The analysis presented in this article confirms that scale-ups are pivotal to Canada’s economic performance. A small cohort of firms creates a disproportionate share of jobs and output growth, justifying the recent policy focus on these firms. The Canadian data show that scale-ups broadly do what is expected: they contribute significantly to employment growth, R&D activity, exporting, and in many cases wages, though short-term productivity effects are less clear. However, the findings also reveal the diversity among scale-ups and some misalignments with policy. Different definitions of scale-ups capture distinct types, highlighting that policy interventions might need to be tailored to these differences (e.g., support those creating jobs versus those pushing the innovation frontier).

No single type of scale-up fulfills all policy objectives. Some scale-ups are champions of job creation but not particularly innovative or productive (such as those in retail or hospitality). Others are at the forefront of innovation and productivity but do not create massive employment (for example, a niche software firm that grows revenue with relatively few highly skilled workers). This typology of employment-intensive versus productivity-intensive scale-ups underscores the need for policymakers to be clear about which outcomes they prioritize. If the goal is innovation and global competitiveness, the focus should be on high-tech or manufacturing scale-ups operating in sectors with high income elasticities of demand and greater R&D and export potential. Current policy rhetoric often bundles employment and innovation goals, assuming that scale-ups will deliver both, but this research suggests that aligning support with the specific contributions and needs of different types of scale-up firms is more effective.

The findings further highlight the degree of sectoral diversity among Canadian scale-ups, challenging the assumption that high growth is predominantly a technology-sector phenomenon. Although the tech sector garners significant attention and support, substantial scale-up potential exists in other areas such as agri-food, clean energy, creative industries, and advanced manufacturing. Policy support should adopt more inclusive, sector-agnostic criteria to leverage growth potential across a broader economic spectrum, ensuring promising scale-ups beyond the tech sectors receive adequate attention and resources. At the same time, our evidence shows that federal innovation support is concentrated in manufacturing, ICT, and professional, scientific, and technical services, and disproportionately benefits larger firms. This concentration raises questions about whether the current mix of support adequately reflects the diversity of scaling pathways documented in the data.

Additionally, while formal R&D activities and export orientation are frequent drivers of successful scaling, the analysis cautions against an overly narrow focus on these pathways. Many scale-ups successfully grow through business model innovations, effective branding, or operational efficiencies that may not be captured under traditional innovation policies

such as the SR&ED tax credit. Expanding policy instruments to recognize and support diverse innovation strategies—beyond formal R&D—could broaden the number and type of firms benefiting from public innovation programs, further enhancing economic growth and innovation performance.

Furthermore, the current support strategy predominantly involves supporting firms after initial scaling – effectively “backing winners.” Although logical from a risk management viewpoint, this approach may limit opportunities to facilitate the initial scaling process for promising firms facing critical financial and operational bottlenecks. International experience highlights significant benefits from proactive interventions to support firms at critical stages of growth transition. Our evidence shows that most program dollars arrive post-scaling, which suggests support is more often amplifying than catalyzing growth. Shifting some policy resources upstream to support firms at earlier growth stages – once they have demonstrated viability but before achieving full scale – could increase the overall scale-up rate and amplify economic impacts.

Overall, the econometric analysis finds significant positive effects of direct federal innovation support (BIGS programs) on scale-up performance across several key metrics, including employment, revenue growth, average wages, R&D intensity, and export activity. Importantly, positive impacts were consistent across various individual BIGS programs, suggesting that, despite differences in specific program objectives or target groups, direct government innovation support broadly and effectively enhances scale-up performance.

While our analysis provides a detailed snapshot of scale-ups in Canada and their interaction with policy, there are limitations to acknowledge. First, our data, being administrative, may under-capture aspects of firm dynamism (e.g., very early-stage startups that haven’t entered official registers, or qualitative factors such as management quality). Second, our support data covers federal programs only and excludes some financing supports delivered through the Business Development Bank of Canada, Export Development Canada, and the Canadian Small Business Financing Program—initiatives that may have broader sectoral

coverage and higher uptake among earlier-stage firms—as well as provincial or local supports, which could be substantial for some firms. Including those could give a fuller picture of the support ecosystem. Third, the impact evaluation, while suggestive, is not a randomized trial – unobservable differences may still bias results. We focused on post-scaling support to mitigate reverse causality, but selection effects are tricky. Further work using techniques such as instrumental variables (e.g., exploiting budget changes in programs, as in Howell (2017), or a regression discontinuity in program eligibility, as in Santoleri et al. (2020)) could strengthen causal claims.

Ultimately, the study underscores the importance of a cohesive policy mix. Scale-up entrepreneurs in Denney et al. (2024) noted that the mix was not fully aligned with their needs. A more aligned mix would provide early-stage support for R&D and commercialization (to seed potential gazelles), growth-stage capital (to scale production/marketing), and complementary measures such as talent attraction (to ease the bringing in of skilled workers) and market development (to help access international opportunities). Canada has pieces of this puzzle in place (SR&ED for R&D, IRAP for early-stage, export programs via the Trade Commissioner Service, etc.), but coordination and smooth transitions among them are lacking (Stewart 2025). Bridging funding gaps and providing advisory support during the scale-up phase (which can be a managerial challenge to navigate hyper-growth) could improve outcomes. Programs like the former Accelerated Growth Service (which coordinated support for high-growth firms across agencies) were promising and could be expanded with a clearer target to support firms that have overcome the scale-up hurdle.

Another avenue for future research is examining scale-up sustainability and longevity. Do Canada’s scale-ups sustain their growth and become major companies (the next BlackBerry or Shopify), or do they often get acquired or fade? Outcomes like acquisitions or IPOs of scale-ups would be interesting to track. Also, understanding the barriers that prevent high-growth firms from continuing to grow (e.g., market saturation, management strain) could inform support beyond the initial scale-up phase. A deeper appreciation of this potential

would require an analysis of those scale-ups operating in industry sectors or market segments with strong income elasticities of demand for their products.

Finally, an important question is how to induce more firms to scale. Are there identifiable predictors in the early stage that a firm will become a scale-up (so that support can be targeted preemptively)? Some researchers (e.g., Guzman and Stern 2017) are developing “nowcasting” models using early indicators (patents, founding team, initial growth) to predict high-growth outcomes. Recent work by Houle and Macdonald (2025) applies similar machine-learning techniques to Canadian data, offering a promising starting point. Further research could build on these methods—refining predictive models, incorporating broader data sources, and testing their effectiveness in guiding programs such as venture capital or accelerator programs. to pick candidates with higher scaling potential.

If Canada is to improve its overall R&D spending-to-GDP ratio, especially in the crucial business sector, increased exports from R&D-performing sectors are key. Since the ill-fated commitment in the 2002 innovation strategy to move Canada from 15th to 5th among OECD countries (Wolfe 2002), governments have repeatedly failed to move the dial on this indicator (Council of Canadian Academies 2018; Stewart 2025). The clear implication of this research is that targeting innovation policy support to the scale-up firms most likely to deliver on improved export performance is essential. By showing how federal innovation policy interacts with scale-ups, this study clarifies both the strengths and the blind spots of Canada’s current approach – and highlights where adjustments could yield the greatest payoff.

Funding

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Appendix

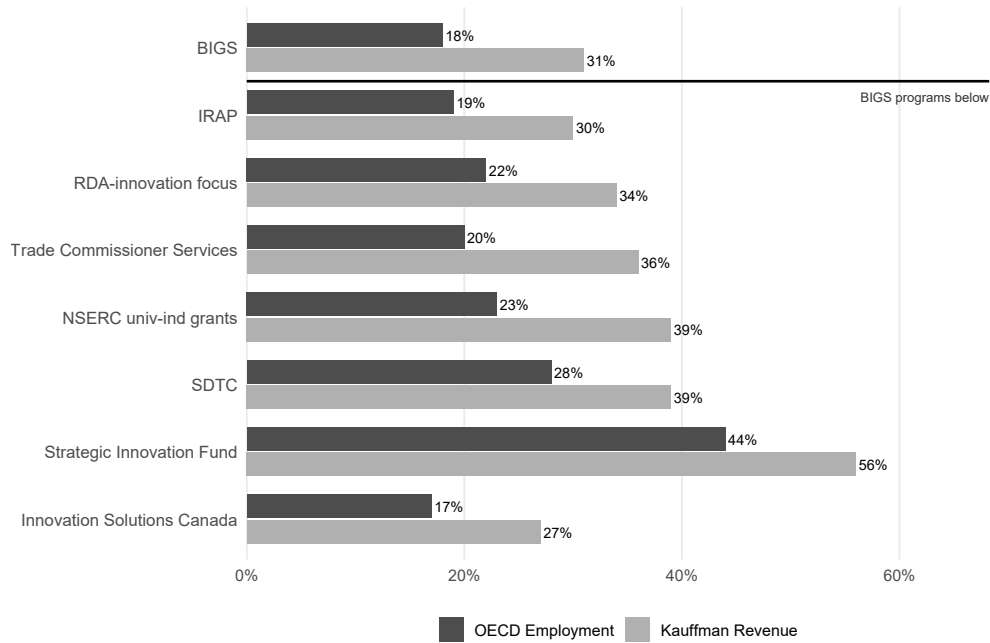


Figure A.1: BIGS Support by Recency of Scaling, OECD and Kauffman Scale-Up Definitions

Note: Figure displays (A) the number of firms receiving BIGS support and (B) the total dollar value of BIGS support, stratified by the time elapsed since the firm's initial scaling episode. Scale-up status is defined using both the OECD employment and Kauffman revenue definitions. Support amounts reflect cumulative BIGS funding received.

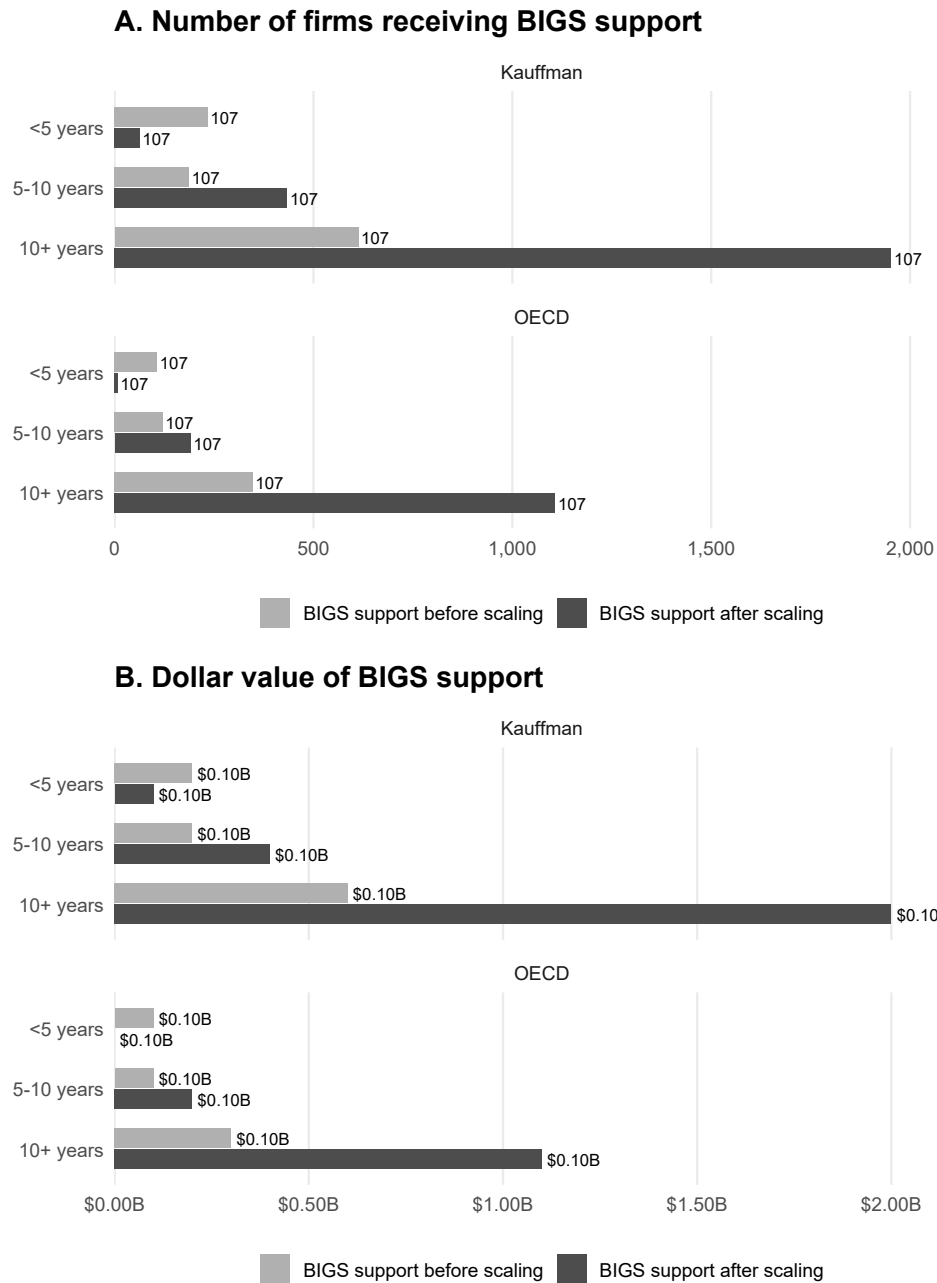


Figure A.2: Program-Specific Support Rates for Scale-Ups, OECD and Kauffman Definitions

Note: Figure reports the share of OECD-employment-defined and Kauffman-revenue-defined scale-ups receiving support from each federal innovation program. Horizontal bars show program-level support rates, with BIGS reported separately from other programs. Dashed or annotated separators indicate program grouping.

Program	OECD Employment			Kauffman Revenue		
	Recipients	Value (\$)	Avg Value (\$)	Recipients	Value (\$)	Avg Value (\$)
IRAP	1,340	106,000,000	79,000	2,145	150,000,000	70,000
RDA–innovation focus	500	223,000,000	446,000	750	302,000,000	403,000
Trade Commissioner Services	1,740	–	–	3,170	–	–
Other programs	1,785	343,000,000	192,000	2,870	579,000,000	202,000
SDTC	40	40,000,000	1,000,000	55	61,000,000	1,109,000
Strategic Innovation Fund	20	65,000,000	3,250,000	25	111,000,000	4,440,000
Innovation Solutions Canada	25	8,000,000	320,000	40	11,000,000	275,000
BIGS	3,675	672,000,000	183,000	6,300	1,031,000,000	164,000
SR&ED	3,465	1,129,000,000	326,000	5,925	1,841,000,000	311,000

Table A.1: Program Support Volume for OECD Employment and Kauffman Revenue Scale-Ups

Note: Table reports counts of supported scale-up firms and the associated total and average dollar value of support received under each federal program. Values represent aggregated support for firms identified as scale-ups under the OECD employment and Kauffman revenue definitions. Dashes indicate missing or inapplicable program-level value reporting.

Outcome	Any BIGS		IRAP		RDA innovation		TCS		NSERC	
	ATT	Sig.	ATT	Sig.	ATT	Sig.	ATT	Sig.	ATT	Sig.
OECD employment										
Employment	10%	***	9%	***	21%	***	14%	***	22%	***
Revenue	17%	***	15%	***	26%	***	24%	***	34%	***
Scaling incidence	5%	***	3%		8%	*	6%	***	8%	**
Average wages	2%	*	4%	***	4%		3%	***	-2%	
R&D expenditures	15%	***	15%	***	13%		12%	***	13%	**
R&D incidence	5%	***	7%	***	0%		6%	***	3%	
Exports	17%	**	6%		46%		12%	*	24%	
Export incidence	4%	***	5%	***	6%	**	7%	***	6%	***
Sales per employee	2%		0%		-6%		4%		-9%	
Kauffman revenue										
Employment	14%	***	14%	***	16%	***	14%	***	20%	***
Revenue	20%	***	18%	***	20%	***	23%	***	27%	***
Scaling incidence	4%	***	4%	***	10%	**	5%	***	5%	**
Average wages	3%	***	3%	**	5%	**	3%	***	2%	
R&D expenditures	14%	***	13%	***	16%	*	14%	***	20%	***
R&D incidence	5%	***	5%	***	1%		5%	***	7%	***
Exports	12%	**	4%		41%	**	14%	***	21%	*
Export incidence	5%	***	5%	***	4%	*	5%	***	7%	***
Sales per employee	1%		0%		-11%	*	3%		3%	

Table A.2: ATTs of BIGS and Specific Program Aggregates on Performance, Matched 2012–2014 Cohort

Note: Each cell reports a fixed-effects estimate of the aggregate treatment effect for the indicated program, separately for OECD-employment-defined and Kauffman-revenue-defined scale-ups. Models include controls for R&D performers, exporters, patent holders, industry–year interactions, and SR&ED support. Treatment and matched control observations are included from $t - 5$ to $t + 5$. Standard errors are clustered at the firm level. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Variable	OECD employment					Kauffman revenue				
	Any BIGS	IRAP	RDA Innov.	TCS	NSERC	Any BIGS	IRAP	RDA Innov.	TCS	NSERC
Before matching										
Employment	***	***	**	***	***	***	***	***	***	***
Revenue	***	***	***	***	***	***	***	**	***	***
Average wages	***	***	***	***	***	***	***	***	***	***
Age	***	**		***		***	***		***	
R&D performer (dummy)	***	***	***	***	***	***	***	***	***	***
R&D expenditures	***	***	***	***	***	***	***	***	***	***
Exports	***	***	***	***	***	***	***	***	***	***
Employment growth (past 3 years)	***	***	*	***	*	***	***	***	***	***
Revenue growth (past 3 years)	***	***	*	***	*	***	***		***	
Years since scaling			***			***			***	
Propensity score	***	***	***	***	***	***	***	***	***	***
After matching										
Employment			**							
Revenue			**							
Average wages	**		**	*					**	
Age										
R&D performer (dummy)										
R&D expenditures			**			***			***	
Exports			*							
Employment growth (past 3 years)		*		**						
Revenue growth (past 3 years)									*	*
Years since scaling (exact match)										
Propensity score										

Table A.3: Matching Diagnostics: Before and After Matching (Kolmogorov–Smirnov Tests)

Note: Each cell indicates whether the Kolmogorov–Smirnov equality-of-distribution test rejects the null hypothesis of equal distributions between treated and matched control firms, before and after matching. Results are shown separately for OECD-employment-defined and Kauffman-revenue-defined scale-ups, and for each program group. Asterisks denote significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.